IES

GenAI for Engineers

One-day Copilot in Excel workshop: SPC, FMEA and DMAIC



Copilot in Excel

SPC • FMEA • DMAIC

ONE DAY | 09:00–17:00

One day, six LP modules

65 minutes of guided Excel practice; demonstrations cover the broader tools.

Morning

Ecosystem · Prompting · Engineering methods

Afternoon

Agents · SharePoint · Security

Practice

Prepared workbook; three short exercises

What Copilot contributes

The engineer verifies the source, formula and interpretation.



Suggest

Tables, formulas and charts

Explain

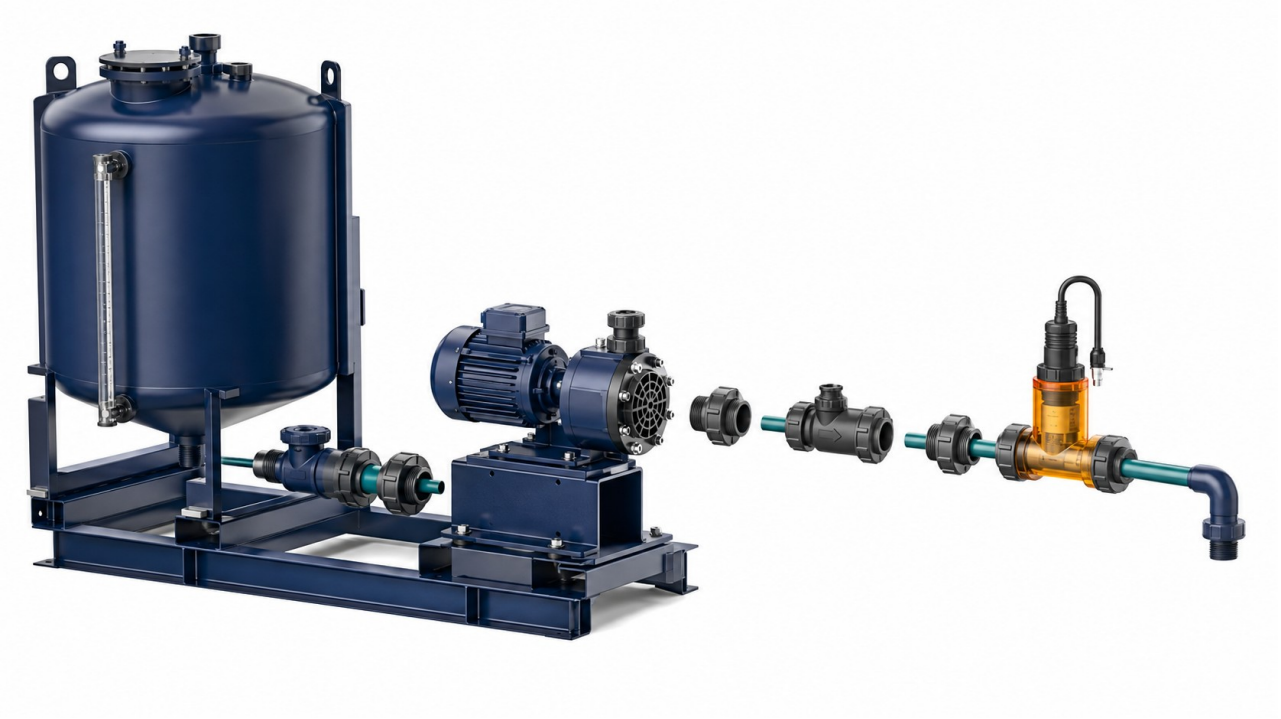
Patterns and evidence gaps

Review

You check calculations and claims

The CW-401 delivery-test case

One dosing skid; linked measurements and maintenance evidence.



Target

62.0 L/h in a standardised test

Comparison

58.9–65.1 L/h fictional specification

Boundary

Not a chemical compliance limit

Prepare Excel for the session

Use approved storage and confirm Copilot access.

Workbook

Save a copy of Worked-Reference.xlsx

Source

Select the relevant named table

Copilot

Use the Excel side pane

Fallback

Record manual formulas honestly

Prompt a workbook operation

Name the table, output and acceptance test.

Role

Engineering data reviewer

Input

Named Excel table and source revision

Steps

Audit, calculate, chart, check

Expected

Formula-based result with limitations

Constraints make the result reviewable

Specify what must be preserved.

Source

RawTests remains unchanged

Data

Units, time order and missingness

Formula

Show rows and parameters

Decision

No invented cause or approval

RISE and E-MAP

Make the task and engineering assumptions explicit.

RISE

Role · Input · Steps · Expectations

E-MAP

Engineering · Metrics · Assumptions · Parameters

Benefit

Less avoidable rework; easier verification

One note becomes a reviewable document

Trainer demonstration: preserve evidence while changing format.

Input

CW-401 test finding and source ID

Prompt

Draft observations, assumptions and review questions

Constraint

Do not invent standards, tolerances or approval

Use three review lenses

Simulated personas expose questions, not independent approval.

01

Reliability

What supports the mechanism?

02

Operations

What is practical and missing?

03

Safety

What consequence needs review?

Debate without forcing agreement

Two short rounds keep the review focused.



Position

Separate evidence-based concerns

Challenge

One specific objection per lens

Response

Cite mode and source record

Moderator

Retain dissent and evidence gaps

Compare the prompt with the result

Better prompting is measured at equal quality.

Fidelity

Source facts preserved

Arithmetic

Formulas independently checked

Uncertainty

No invented cause or approval

Effort

Draft plus review and correction

Methodology primer · 15 minutes

Understand the output before asking Copilot to interpret it.

SPC

Variation over time; signal versus specification

FMEA

Mode, effect, cause, control; S × O × D

DMAIC

Define · Measure · Analyse · Improve · Control

Three guided exercises · 65 minutes

Use the prepared worked workbook; finish one small output per exercise.

11:45–12:10

SPC: one verified chart interpretation

12:10–12:30

FMEA: two checked rows and an evidence gap

12:30–12:50

DMAIC: rates, limitation and next evidence

12:50–13:00

Shared review; stop for lunch

SPC3 · Plot and interpret

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Review the existing Analysis and SPC_Summary chart. Keep the prepared formulas and fixed baseline limits unchanged. Identify valid monitoring points outside the limits and check one example against the source row. Return only three sentences: observed signal, supporting evidence, and an unresolved limitation. Do not rebuild the chart or calculate capability.

Input

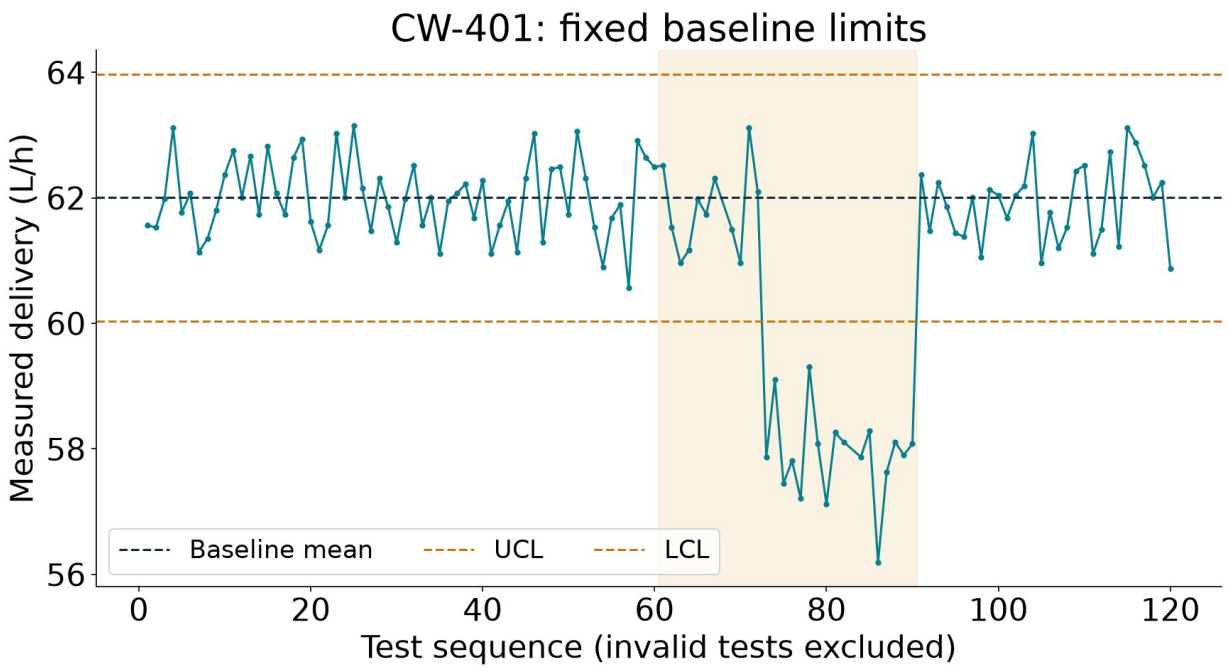
Select the named source table

Review

Check the changed cells, formulas and claims

The monitoring period shifts

Fixed limits reveal a change that requires investigation.



LCL

60.03757 L/h

UCL

63.96743 L/h

Signal

17 monitoring points below LCL

Specifications and control limits differ

Requirements and observed behaviour answer different questions.

01

Specification

58.9–65.1 L/h for this teaching test

02

Control limits

Estimated from the chosen baseline

03

Interpretation

Neither alone establishes physical cause

FMEA2 · Calculate and prioritise

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Review the prepared FMEA_Review. Check the RPN formulas for FM01 and FM04, preserving mode IDs and provisional scores. Explain why the severity-9 row still needs review. Return one evidence gap and one proposed action for FM01; keep residual ratings unassessed. Do not rebuild the table or chart.

Input

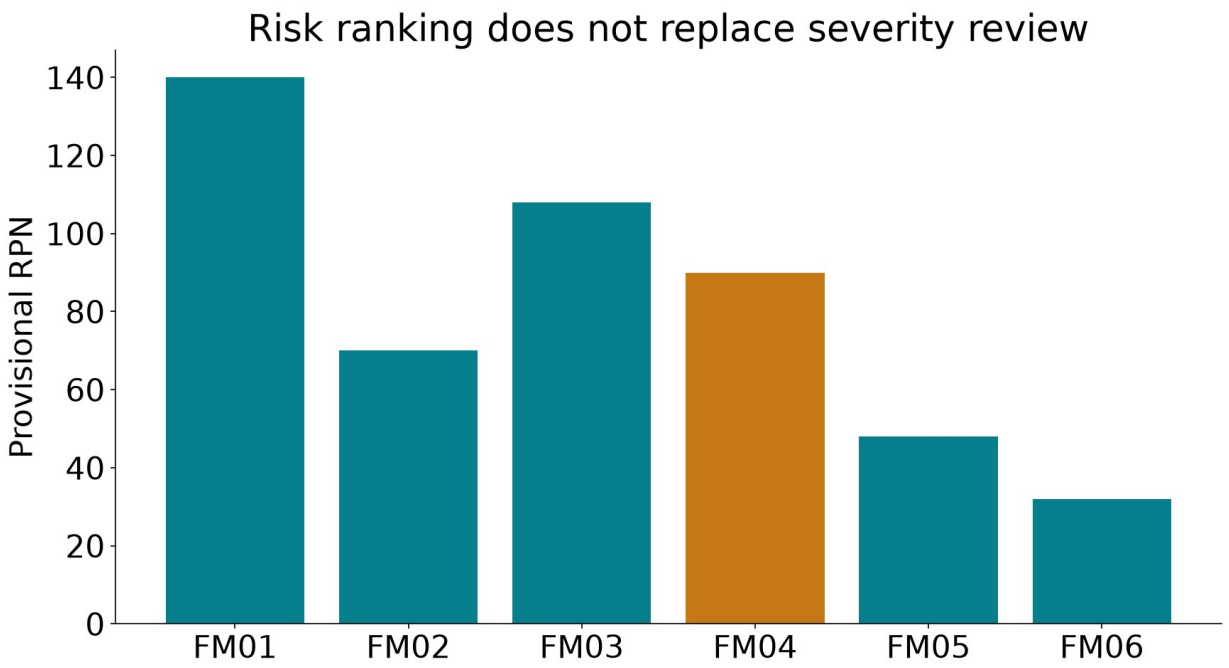
Select the named source table

Review

Check the changed cells, formulas and claims

Severity review can outrank RPN

FM04 has S = 9 and still needs explicit review.



FM01

Highest RPN 140

FM03

RPN 108

FM04

RPN 90; severity 9

DMAIC1 · Define and measure

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Review the prepared Rate_Comparison, WeeklyEvents and DMAIC tables. Check baseline and pilot rates using summed deviations divided by summed opportunities. Return three cells: measured change, limitation and next evidence. Do not create a full report or a new chart, and do not claim a causal effect from the short pilot.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

Use rates when periods differ

The eight-week log is separate from the120-test investigation.

Baseline

93 /1008 =9.22619%

Pilot

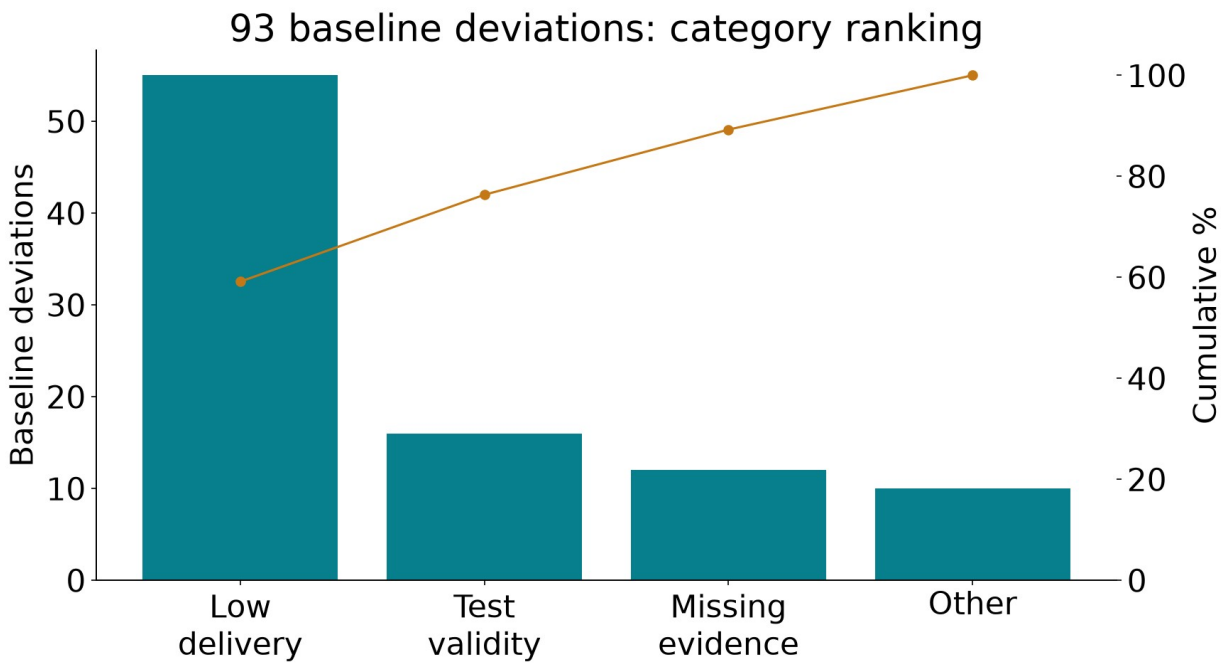
11 /336 =3.27381%

Scope

Six baseline weeks; two pilot weeks

Pareto directs attention

Category counts prioritise investigation; they do not prove causes.



Total

93 baseline deviations

Top category

Low delivery 55

Top two

71 /93 =76.34%

Stop at the agreed output

Rebuilding the full workbook is optional after class.

SPC

Three-sentence chart interpretation

FMEA

One evidence gap and action proposal

DMAIC

Three-cell decision summary

Agents · trainer demonstration

Participants observe a prepared assistant; no agent-building lab.

Instructions

Bounded role, inputs and output format

Knowledge

Approved template and rating scales

Test

One case, one failure and one correction

Reuse a reviewed workflow

The trainer shows the FMEA and report patterns.

Input

Case facts and approved template

Transform

Populate fields and identify gaps

Review

Source checks and human release decision

Research and analysis answer different questions

Compare source quality and arithmetic separately.

Research

Which source supports the comparison?

Analysis

Which table and formula support the result?

Check

Open citations; recompute a selected value

From minutes to a reviewed report

A workflow still needs ownership and evidence.

Capture

Decision, action, owner and due date

Draft

Map actions into a report template

Release

Reviewer checks status and unresolved items

SharePoint grounding · demonstration

Follow one answer back to an approved document.

Retrieve

Find a passage in the scoped library

Cite

Document, revision and location

Verify

Open the source and check applicability

Permissions define the knowledge boundary

Demonstrate access and discuss governance; no site-building task.

Scope

Approved training library

Access

User permissions and document controls

Boundary

Retrieved instructions do not override the task

Seven checks before using an AI output

Apply the framework to one displayed engineering claim.

1–2

Classify data; check governance and access

3–4

Prompt securely; verify outputs

5–6

Human sign-off; source and prompt audit trail

7

Recognise and report suspicious content

Keep a small engineering AI toolkit

One reusable prompt and one review rule are enough today.

Prompt

Named evidence, explicit constraints, expected output

Review rule

Recompute one value and open one source

After class

Optional full workbook rebuild

Closing · 16:45–17:00

One result, one limitation and one next step.

Recap

SPC · FMEA · DMAIC

Judgement

What remains unproven?

Next step

Choose one bounded workplace trial